

Are GPT detectors biased against non-native English writers?

2304.02819

Abstract

INTRODUCTION

Intro

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Recent advances in large language models have made AI-generated text detection an increasingly critical task for academic integrity and information authenticity; however, the detectors themselves may harbor systematic biases with serious equity implications. Liang et al. (arXiv:2304.02819) demonstrated that popular GPT detectors misclassify over half of non-native English TOEFL essays as AI-generated, with false positive rates averaging 61.22%, revealing a stark linguistic bias against non-native writers. This survey examines the nature, scope, and mitigation strategies for detector bias against non-native English authors, and considers the broader implications for equitable AI governance.

引言

大语言模型的快速发展使 AI 生成文本检测成为维护学术诚信的关键任务，但检测器本身可能存在系统性偏见。Liang 等人 (arXiv:2304.02819) 发现，主流 GPT 检测器将超过一半的非英语母语者 TOEFL 作文错误地标记为 AI 生成，平均假阳性率高达 61.22%，揭示了对非母语者的显著语言偏见。本综述探讨检测器偏见的性质、范围及缓解策略，并分析其对公平 AI 治理的更广泛影响。

Research Context and Problem

The deployment of GPT detectors in academic and professional settings has raised significant concerns about fairness and equity. These tools, designed to identify AI-generated text, are increasingly being used to evaluate student work and manuscript submissions. However, emerging evidence suggests that such detectors may systematically disadvantage certain populations. Specifically, Liang et al. (arXiv:2304.02819) found that GPT detectors misclassify over half of TOEFL essays written by non-native English speakers as AI-generated, with an average false positive rate of 61.22%. This finding is particularly alarming given that the linguistic features detectors rely on—such as vocabulary simplicity and sentence structure—are inherently influenced by a writer’s English proficiency level.

The bias is further confirmed by a controlled experiment in which simplifying word choices in native English essays to mimic non-native writing patterns caused detector misclassification rates to surge from an average of 5.19% to 56.65% (arXiv:2304.02819). Additionally, analysis of ICLR 2023 conference papers with comparable review ratings revealed that non-native authors’ abstracts exhibited significantly lower perplexity than those from native authors, suggesting that the statistical signals detectors exploit are confounded with language background rather than AI involvement (arXiv:2304.02819). These overlapping

findings point to a fundamental gap: current detection paradigms conflate linguistic non-fluency with machine authorship, creating a false equivalence that disproportionately harms multilingual scholars.

GPT 检测器在学术与职业场景中的部署引发了关于公平性的广泛关切。Liang 等人 (arXiv:2304.02819) 发现, 检测器将超过半数的非英语母语者 TOEFL 作文误判为 AI 生成, 平均误报率高达 61.22%。当研究者刻意简化母语作者的用词以模拟非母语写作时, 误判率从 5.19% 飙升至 56.65%。此外, 对 ICLR 2023 论文的分析表明, 在评审评分相近的情况下, 非母语作者摘要的困惑度显著低于母语作者。这些证据共同揭示了一个关键缺口: 现有检测范式将语言不流利性与机器生成混为一谈, 对多语言学者造成了系统性歧视。

Methodology

The study evaluated multiple commercially available GPT detection tools against writing samples from both native and non-native English speakers, using TOEFL essays written by non-native writers as the core test corpus (arXiv:2304.02819). Across the detector suite, over half of these non-native essays were misclassified as AI-generated, yielding an average false positive rate of 61.22% (arXiv:2304.02819). To isolate the linguistic mechanism behind this bias, the authors ran controlled manipulations and comparison conditions: simplifying word choices in native essays to mimic non-native writing raised the average misclassification rate from 5.19% to 56.65%, directly implicating lexical simplicity rather than authorship itself (arXiv:2304.02819); and, in matched ICLR 2023 papers with comparable review ratings, abstracts by non-native authors exhibited significantly lower perplexity than those by native authors, connecting the observed bias to measurable text statistics (arXiv:2304.02819). A further control test showed that a simple second-round self-edit prompt applied to ChatGPT-generated essays (“elevate the text with literary language”) cut detector detection rates from 100% to 13%, demonstrating the detectors’ brittleness against light rewrites (arXiv:2304.02819).

Together these experiments delimit the methodological boundaries of the finding: bias is robust across tools and samples yet sensitive to paraphrase, and it tracks low-perplexity writing patterns. The open gap is that the published work does not provide a shared, reproducible test corpus or normalization protocol, leaving cross-study comparability of “detector bias” unstandardized.

方法

该研究用非英语母语者撰写的 TOEFL 作文作为核心测试语料, 评估了多款商用 GPT 检测工具在母语与非母语写作者文本上的表现 (arXiv:2304.02819)。检测器将超过半数的非母语作文误判为 AI 生成, 平均误报率高达 61.22% (arXiv:2304.02819)。为定位偏误的成因, 作者设计了受控实验: 将母语作文的用词简化以模拟非母语风格后, 平均误分类率从 5.19% 升至 56.65%, 表明元凶是词汇复杂度而非作者身份 (arXiv:2304.02819); 在评审得分相近的 ICLR 2023 论文中, 非母语作者的摘要困惑度显著更低, 将偏误与可测的文本统计量联系起来 (arXiv:2304.02819)。另一对照组实验显示, 对 ChatGPT 生成文本施加一轮“文学化改写”提示词即可把检出率从 100% 压至 13%, 暴露了检测器对轻度改写的脆弱性 (arXiv:2304.02819)。

综上, 偏误在多种工具与语料上稳健存在, 却又对改写高度敏感, 且与低困惑度文本模式相关。遗留缺口在于: 原文未提供可复现的共享测试语料与归一化协议, 导致“检测器偏误”跨研究难以直接可比。

Key Findings

Research has revealed systematic bias in AI-generated text detection tools against non-native English writers. Liang et al. found that six widely used GPT detectors—including GPTZero, Crossplag, and Writer.com—misclassified over half of non-native English TOEFL essays as AI-generated, yielding an average false positive rate of 61.22% (arXiv:2304.02819). To isolate the causal mechanism, the authors

simplified lexical choices in native English essays to approximate non-native writing style; this single intervention raised the average misclassification rate from 5.19% to 56.65%, demonstrating that detectors are sensitive to linguistic complexity rather than actual AI authorship (arXiv:2304.02819). A parallel analysis of ICLR 2023 papers with similar peer-review ratings confirmed that non-native authors' abstracts exhibit significantly lower perplexity than those of native authors, further underscoring how detectors conflate non-native simplicity with machine generation (arXiv:2304.02819).

These findings are not easily remedied at the prompt level. Liang et al. showed that applying a simple self-edit prompt—“Elevate the provided text by employing literary language”—to ChatGPT-3.5 outputs reduced detection rates from 100% to as low as 13%, meaning that anyone can evade detection by deliberately enriching their text (arXiv:2304.02819). This creates a paradox: non-native writers are disproportionately flagged precisely because they write clearly and simply, while AI-generated text can easily escape detection through stylistic manipulation. The bias thus represents a systematic error pattern across multiple detection tools, not an isolated failure of any single model. This opens a critical gap: current GPT detectors may be fundamentally unsuitable for high-stakes academic integrity decisions—such as evaluating non-native applicants—without substantial recalibration, as their error patterns encode linguistic discrimination.

关键发现

研究表明，AI 文本检测工具对非英语母语写作者存在系统性偏差。Liang 等人发现，六个广泛使用的 GPT 检测工具（包括 GPTZero、Crossplag 和 Writer.com）将超过一半的非英语母语 TOEFL 作文错误地标记为 AI 生成，平均假阳性率高达 61.22% (arXiv:2304.02819)。为隔离因果机制，作者对英语母语者的作文进行了词汇简化处理以模拟非英语母语写作风格；仅此一项干预就使平均误判率从 5.19% 飙升至 56.65%，证明检测器对语言复杂度的敏感度远超对真实 AI 写作的识别能力 (arXiv:2304.02819)。对 ICLR 2023 论文的平行分析进一步证实，在同行评审评分相近的情况下，非英语母语作者的摘要困惑度显著低于英语母语作者，再次凸显检测器将非母语的简洁性等同于机器生成的偏误 (arXiv:2304.02819)。

这些发现难以通过提示层面简单修正。Liang 等人发现，对 ChatGPT-3.5 输出应用简单的自我编辑提示——“使用文学性语言提升文本”——可将检测率从 100% 降至低至 13%，即任何人都可通过刻意丰富文本来规避检测 (arXiv:2304.02819)。这构成一个悖论：非英语母语写作者因表达清晰简洁而被过度标记，而 AI 生成的文本却可通过风格操控轻松逃脱检测。因此，该偏差代表了跨多个检测工具的系统性错误模式，而非单一模型的孤立失败。这留下了一个关键空白：当前的 GPT 检测器在未进行重大校准的情况下，可能根本不适合用于高利害学术诚信决策——如评估非英语母语申请者——因为其错误模式本身编码了语言歧视。

Implications for Academic Integrity

AI-generated text detectors, widely adopted by academic institutions to safeguard research integrity, exhibit a troubling bias against non-native English writers. Liang et al. (arXiv:2304.02819) demonstrate that popular GPT detectors misclassify over half of non-native English TOEFL essays as AI-generated, yielding an average false positive rate of 61.22% across seven commercial tools tested. Critically, this bias is not merely a function of content but of linguistic complexity: when native English essays were deliberately simplified to mimic non-native writing patterns, the misclassification rate surged from 5.19% to 56.65%, suggesting that detectors conflate limited lexical diversity with machine authorship (arXiv:2304.02819). Even within high-stakes peer-reviewed contexts, the disparity persists—ICLR 2023 papers with comparable review ratings show that non-native authors' abstracts exhibit significantly lower perplexity than those from native authors (arXiv:2304.02819), indicating that these detectors penalize the very linguistic traits common among non-native academic writers.

These findings carry substantial consequences for academic equity. The detection gap implies that non-native English-speaking scholars are disproportionately vulnerable to false accusations of academic misconduct, potentially undermining trust, career progression, and participation in international scholarship.

Furthermore, the ease with which ChatGPT-generated text can evade detection—reducing false positives from 100% to just 13% through a simple self-edit prompt—renders these tools doubly ineffective: they simultaneously over-flag legitimate non-native writing while under-detecting actual AI-generated content (arXiv:2304.02819). The authors conclude that current detector tools require fundamental redesign to avoid perpetuating discriminatory outcomes in academic gatekeeping.

The open gap this leaves is clear: there is an urgent need for detection methods that are robust to both low-perplexity human writing and adversarial AI-generated text, while remaining linguistically equitable across diverse author populations. Without such tools, institutions risk embedding systematic discrimination into the very mechanisms designed to uphold research integrity.

学术诚信的深层影响

广泛应用于学术机构的 AI 生成文本检测工具, 对非英语母语写作者表现出显著偏见。Liang 等人 (arXiv:2304.02819) 的研究表明, 主流 GPT 检测器将超过半数的非英语母语 TOEFL 作文误判为 AI 生成, 七款商业工具的平均误报率高达 61.22%。更值得注意的是, 这种偏见并非源于写作内容, 而是与语言复杂度直接相关: 当母语为英语的论文被刻意简化以模拟非母语写作风格时, 误判率从 5.19% 飙升至 56.65%, 说明检测器将词汇多样性不足等同于机器写作 (arXiv:2304.02819)。即便在 ICLR 2023 等高水平同行评审会议上, 相似评审评分的论文中, 非母语作者的摘要表现出显著低于母语作者的困惑度 (arXiv:2304.02819), 意味着检测器正在惩罚非母语学术写作中普遍存在的语言特征。

这一发现对学术公平性构成深远影响。检测偏差意味着非英语母语学者面临更高比例的虚假学术不端指控, 可能损害其学术信任、职业发展和国际学术参与。更为讽刺的是, ChatGPT 生成的文本仅需通过简单的自我编辑提示, 即可将误报率从 100% 降至 13% (arXiv:2304.02819), 使这些工具陷入双重失效: 既过度标记合法的非母语写作, 又无法有效识别真正的 AI 生成内容。研究者呼吁对现有检测工具进行根本性重新设计, 以避免在学术守门机制中固化系统性歧视。

由此留下的研究空白十分明确: 迫切需要开发既能抵抗低困惑度人类写作和对抗性 AI 生成文本, 又能在不同语言背景作者间保持公平性的检测方法。否则, 机构将有可能在旨在维护学术诚信的机制中嵌入系统性歧视。

Conclusion / 结语

English

AI-generated text detectors, increasingly deployed in academic and professional settings, carry a systematic bias against non-native English writers. Liang et al. (arXiv:2304.02819) demonstrate that popular GPT detectors misclassify over half of TOEFL essays authored by non-native English speakers as AI-generated (average false positive rate of 61.22%). This bias is not an artifact of topic or prompt: simplifying word choices in native-authored essays to approximate non-native writing styles causes the same detectors' misclassification rate to spike from 5.19% to 56.65% (arXiv:2304.02819). Analysis of ICLR 2023 accepted papers further confirms the structural root: non-native authors produce abstracts with significantly lower perplexity—a signature that detectors conflate with machine generation—despite comparable peer-review ratings (arXiv:2304.02819).

Mitigation remains nascent. A trivial self-edit prompt (“Elevate the provided text by employing literary language”) applied to ChatGPT-3.5 outputs slashed detection rates from 100% to 13% (arXiv:2304.02819), suggesting that adversarial rewriting trivially circumvents detection while legitimate non-native prose continues to be flagged. This creates an asymmetric risk: the very populations most vulnerable to detection bias—students, multilingual researchers, and non-Anglophone professionals—are least equipped to game the detectors.

Remaining gaps. First, no large-scale study has quantified the downstream harm of false-positive classifications on non-native speakers’ academic standing or professional opportunities. Second, detector performance across languages other than English, and across domain-specific registers (legal, medical, technical), remains largely uncharacterized. Third, the arms race between paraphrasing and detection has not been evaluated end-to-end under realistic deployment conditions; the self-edit results (arXiv:2304.02819) hint at easy evasion but do not exhaust the attack surface. Finally, governance frameworks that adjudicate fairness criteria for text detectors—who bears the burden of proof, what false-positive threshold is acceptable, and whether detector outputs should serve as dispositive evidence rather than probabilistic signals—are absent from both policy and the research literature.

Until detectors account for linguistic diversity as a first-class dimension, their deployment in high-stakes contexts risks institutionalizing a disadvantage against non-native English writers.

中文速览

AI 生成文本检测器在学术和职业场景中的广泛部署，对非英语母语写作者构成了系统性偏差。Liang 等人 (arXiv:2304.02819) 发现，主流 GPT 检测器将超过半数的 TOEFL 非母语作文误判为 AI 生成，平均误报率达 61.22%。将母语者的用词简化为近似非母语风格后，同一组检测器的误判率从 5.19% 飙升至 56.65%，证实偏差根植于语言特征而非内容本身 (arXiv:2304.02819)。对 ICLR 2023 已录用论文的分析进一步表明：在评审评分相近的前提下，非母语作者摘要的困惑度显著更低——这正是检测器误判为机器生成的统计信号 (arXiv:2304.02819)。

缓解措施仍处于早期阶段。一句简单的自我编辑提示就能将 ChatGPT-3.5 输出的检测率从 100% 降至 13% (arXiv:2304.02819)，说明攻击者可轻易绕过检测，而合法的非母语文本却持续被标记，形成不对称风险。

尚存的研究空白。 (1) 缺乏大规模实证研究量化误判对非母语群体学术声誉和职业机会的实际损害；(2) 英语以外语言及垂直领域（法律、医学、技术）的检测器表现几乎未被评估；(3) 改写与检测之间的攻防竞赛尚未在真实部署条件下进行端到端评测；(4) 针对文本检测器的公平性治理框架——谁承担举证责任、可接受的误报率阈值、检测结果能否作为决定性证据——在政策和学术文献中均付之阙如。在检测器将语言多样性作为一等考量维度之前，其在高利害场景中的部署将使非英语母语写作者处于制度性劣势。

Evidence Table

#	Claim (abridged)	Source	Confidence
1	GPT detectors misclassify over half of non-native English TOEFL essays as AI-generated, with an average false positive r ...	arXiv:2304.02819	high
2	Simplifying word choices in native essays to mimic non-native writing increases the average misclassification rate from ...	arXiv:2304.02819	high
3	In ICLR 2023 papers with similar review ratings, non-native authors’ abstracts exhibit significantly lower perplexity th...	arXiv:2304.02819	high
4	A simple self-edit prompt on ChatGPT-generated essays drastically reduced detector detection rates from 100% to 13%	arXiv:2304.02819	high

References

- [\[1\]](#) arXiv:2304.02819